

Splitting, Stickerballs, and Flag Complexes for Taut Fillings

What the Lean development `lean/Taut` proves

This document states, as ordinary mathematics, *exactly* what the Lean development `lean/Taut` proves about taut fillings — no strengthening inferred from prose. The results are:

- (1) taut fillings of almost-disjoint cycles *split*, and their filling volumes *add* (§2), with a rational analogue (Corollary 2.5);
- (2) a taut filling of a combinatorial 2-sphere is automatically simplicial, pure, and triangle-bounded, and forms an *AnyRootedStickerball* — a clean-shellable complex that can be shelled starting from any tetrahedron (§3);
- (3) the same filling is a *flag complex* (§4).

Almost-disjoint splitting is the engine; the stickerball and flag-complex theorems are inductions on the size of the filling driven by it. Typewriter names are the exact Lean identifiers, listed with their status in §5; everything labelled a theorem or lemma is fully formalized with no `sorry` and no project axioms. Where a notion is weaker than its classical namesake, or a classical bridge is *not* formalized, this is stated explicitly and flagged in the text as “not Lean-proven here” (§5–6).

1 Definitions

Fix a linearly ordered vertex set V . A simplex is a finite subset $s \subseteq V$; if $|s| = k + 1$ we call it a k -simplex. A *chain* is a finitely supported function M from simplices to $\mathbb{Z}$ (`Chain V`), written $M = \sum_s M(s) s$, with orientations fixed by the vertex order. The boundary ∂ is the usual signed face operator (`bdry`); $\partial\partial = 0$. The *norm* is $\|M\| = \sum_s |M(s)|$ (`norm`); a chain is *simplicial* if every coefficient lies in $\{-1, 0, 1\}$ (`SimplicialChain`), and then $\|M\|$ is the number of simplices in its support. (For a taut filling of a 2-sphere, simpliciality is automatic — Theorem 3.1 — and is nowhere assumed.)

Definition 1.1 (Filling volume; taut filling). The *filling volume* of a chain X is $\text{Zvol}(X) = \inf\{\|M\| : \partial M = X\}$ (`Zvol`, literally an inf over $\mathbb{N}$). For a cycle ($\partial X = 0$) the cone over any vertex is a filling (`bdry_cone_of_closed`), so the inf is attained and

$$\text{Zvol}(X) = \min\{\|M\| : \partial M = X\} \quad (\text{exists_fill_eq_Zvol}).$$

(`Zvol` is defined for every chain; it is used below only on cycles.) A chain M is a *taut filling* of X if $\partial M = X$ and $\|M\| = \text{Zvol}(X)$, i.e. M is a norm-minimal filling of its own boundary (`IsTaut`).

Tautness contributes exactly three things to the proofs, all consequences of minimality:

(T1) *Boundary control*: $\partial M = X$ holds on the nose.

- (T2) *Purity / no junk*: the generators of a taut filling of an n -cycle are all $(n+1)$ -simplices (`IsTaut.dim_pure`), and every vertex of M lies on the boundary (`IsTaut.vert_subset`, via `IsTaut.no_internal_vertex`).
- (T3) *No extraneous cancellation*: a taut chain contains no *nontrivial* complete cone $\text{cone}_x W$ as a subchain — W closed ($\partial W = 0$), the apex x absent from W ($\deg_x W = 0$), and some vertex x' present in W ($\deg_{x'} W \neq 0$) (`not_taut_complete_cone`); such a cone could be refilled with strictly fewer simplices.

“Taut” is therefore purely combinatorial ℓ^1 -minimality with these structural corollaries; it is not a metric or curvature condition.

Definition 1.2 (Combinatorial 2-sphere). A set σ of triangles is a *combinatorial 2-sphere* (`IsSphere2`) if it is pure 2-dimensional, every edge lies in exactly two triangles, every vertex link is *connected*, the 1-skeleton is connected, and $V + F = E + 2$ (that each vertex link is in fact a cycle, i.e. 2-regular, is then derived: `link_two_regular`). These properties are taken as the definition.

Definition 1.3 (AlmostDisjoint). Fix $n \geq 1$ and $A, B \subseteq V$. Closed pure n -cycles X, Y are *Almost-Disjoint over* (A, B) if every generator of X is an $(n+1)$ -subset of A , every generator of Y is an $(n+1)$ -subset of B , $\partial X = \partial Y = 0$, and

$$|A \cap B| \leq n + 1.$$

Mathematically: the two regions overlap in at most an n -simplex’s worth of vertices. This is the *entire* separation hypothesis — no metric, no topological transversality. (Informal name; in Lean it is the hypothesis bundle of `Zvol_add_of_almost_disjoint_full` and `IsTaut.splits_full`.)

For the stickerball theorem we build complexes of tetrahedra one at a time. Let τ be a set of tetrahedra (4-element simplices) with current boundary B (a set of triangles).

Definition 1.4 (Clean glue step; clean shelling). Adding a tetrahedron t to τ is a *clean glue step* (`CleanGlueStep`) if:

- (a) t meets the current boundary B in one or two of its four triangular faces, and the new boundary is B updated by symmetric difference with the faces of t ;
- (b) $t \notin \tau$ and no already-interior face of t is rogue (each old face of t lies in a shared boundary triangle); and the three *link certificates*:
- (c1) each triangle of t lies in at most one tetrahedron of τ (so at most two after adding t);
- (c2) each edge of t is new to τ , or shares an apex with τ ’s edge-link there;
- (c3) each vertex of t is new to τ , or shares an apex with τ ’s vertex-link there.

A *clean shelling* of τ with boundary B is an ordering $t_1, \dots, t_k$ of τ in which t_1 is a single tetrahedron and each t_{i+1} is added to $\{t_1, \dots, t_i\}$ by a clean glue step, terminating at boundary B (`IsCleanShelling`, `CleanShellFrom`).

The certificates (c1)–(c3) are precisely the hypotheses needed to preserve, across one insertion, the three conditions defining a *clean 3-complex*.

Definition 1.5 (Clean 3-complex). A set τ of tetrahedra is a *clean 3-complex* (`IsClean3Complex`) if it is pure (3-dimensional), triangle-bounded (every triangle in at most two tetrahedra), and *normal*: both its edge-links and its vertex-links are connected.

Definition 1.6 (Stickerball; root; RootedStickerball; AnyRootedStickerball). A set τ of tetrahedra is a *stickerball* with boundary B if it admits a clean shelling ending at B and $B \neq \emptyset$ (`IsStickerball`). A *root* of a stickerball is a tetrahedron $t \in \tau$ at which some clean shelling can *begin*. “Rooted at t ” (*RootedStickerball*) is not a standalone Lean predicate; it names the existence clause for that t inside `FreelyCleanShellable`. The stickerball is an *AnyRootedStickerball* if every tetrahedron of τ is a root: in Lean, `IsAnyrootedStickerball` = `IsStickerball` together with `FreelyCleanShellable`, where `FreelyCleanShellable` asserts a clean shelling beginning at each $t \in \tau$.

Remark. “RootedStickerball” and “AnyRootedStickerball” are stated in Lean-backed terms. The clean shelling certifies normality (§6); we do not silently upgrade “stickerball” to “shellable ball” or “3-ball” — the proved bridge and the missing bridge are stated in §6.

2 Splitting under AlmostDisjoint

The engine of the splitting theorem is a small-support triviality (Lemma 2.2) together with the no-cone property (T3) of taut chains. We also record here, once, the one homological input — $H_1 = 0$ and the cut it yields (Lemma 2.1); the splitting theorem does *not* use it (its analogue of closedness is the coefficient-level `bdry_lk_eq_zero`, not the cut), but the sphere-cutting reductions of §3 do.

The homological input (used in §3)

Lemma 2.1 ($H_1 = 0$ and the cut). *On a combinatorial 2-sphere, over $\mathbb{F}_2$, every 1-cycle is a boundary: $\text{im } \partial_2 = \ker \partial_1$ (`range_bd2_eq_ker_bd1`). Consequently, if γ is a triple of vertices whose three edges lie in σ but which is not a face of σ , then the 1-cycle of its edges bounds a set W of faces of σ (`exists_cut`).*

Proof. The $\mathbb{F}_2$ 2-cycles are spanned by the all-faces chain (dual-graph connectivity), so $\dim \text{im } \partial_2 = F - 1$; the connected 1-skeleton gives $\dim \text{im } \partial_1 = V - 1$, hence $\dim \ker \partial_1 = E - (V - 1)$, which equals $F - 1$ by $V + F = E + 2$. With $\text{im } \partial_2 \subseteq \ker \partial_1$ this forces equality. A non-face triple γ has each vertex on exactly two of its edges, so its edge-chain is a 1-cycle and therefore a boundary $\partial_2 W$. $\square$

Lemma 2.2 (Small-support triviality). *A closed chain whose generators all have $n + 1$ vertices, supported on a set of at most $n + 1$ vertices, is zero: with fewer than $n + 1$ vertices there is no $(n+1)$ -generator at all (`eq_zero_of_supp_card_lt`, closedness not needed), and on exactly $n + 1$ vertices the single possible generator is killed by closedness (`eq_zero_of_closed_supp_card_eq`).*

Proof. On at most $n + 1$ vertices there is at most one $(n+1)$ -subset, so the chain is $c \cdot s$ for a single simplex; $\partial(cs) = 0$ forces $c = 0$. $\square$

The split

Fix a common vertex $p \in A \cap B$. There is a norm-non-increasing *kill operator* $K_{A,p}$ (`Projection.lean`, a cone projection toward p) with: $K_{A,p}$ fixes every chain confined to A (`Kmap_eq_self`); and $K_{A,p}$ annihilates every closed pure n -cycle confined to B (`Kmap_eq_zero_of_closed`), because its image is a closed pure cycle on $A \cap B$, hence on $\leq n + 1$ vertices, hence zero by Lemma 2.2. Symmetrically for $K_{B,q}$ with $q \in A \cap B$, $q \neq p$.

Theorem 2.3 (Additivity). *If X, Y are AlmostDisjoint over (A, B) with $n \geq 1$, then $\text{Zvol}(X+Y) = \text{Zvol}(X) + \text{Zvol}(Y)$ (`Zvol_add_of_almost_disjoint_full`).*

Proof. ($\leq$) The sum of minimal fillings of X and Y is a filling of $X+Y$, so by subadditivity of the norm, $\text{Zvol}(X+Y) \leq \|M_X\| + \|M_Y\| = \text{Zvol } X + \text{Zvol } Y$ (`Zvol_le, nrm_add_le`). ($\geq$) Let M be a minimal (hence taut) filling of $X+Y$, so $\text{Zvol}(X+Y) = \|M\|$; its generators are pure $(n+2)$ -simplices lying in $A \cup B$ (`IsTaut.dim_pure, IsTaut.vert_subset`). Each generator t splits as $(t \cap A \cap B) \sqcup (t \setminus A) \sqcup (t \setminus B)$; as $|A \cap B| \leq n+1 < n+2$, t cannot avoid both $t \setminus A$ and $t \setminus B$, so t is killed by *at least one* of $K_{A,p}, K_{B,q}$ (`Kkills_or_Kkills`, an inclusive “or”). Equivalently the two retained sets are disjoint — a generator may be killed by both (retained by neither), e.g. one with $|t \setminus A| = |t \setminus B| = 2$, and that is harmless. With the per-map bound $\|K_{A,p}M\| + \|M \upharpoonright_{\text{kill}_A}\| \leq \|M\|$ (`nrm_kmap_add_killed_le`) and $\|M\| \leq \|M \upharpoonright_{\text{kill}_A}\| + \|M \upharpoonright_{\text{kill}_B}\|$ (`nrm_le_killed_add_killed`) this gives $\|K_{A,p}M\| + \|K_{B,q}M\| \leq \|M\|$. Because $K_{A,p}$ fixes the A -part of ∂M and kills its closed B -part, $K_{A,p}M$ fills X and $K_{B,q}M$ fills Y ; thus $\|M\| \geq \text{Zvol } X + \text{Zvol } Y$. (Exclusivity — each generator killed by *exactly* one — is not used here; it holds only later, for the taut filling in Theorem 2.4, once `no_double_kill` rules out double kills.) $\square$

Theorem 2.4 (Splitting under AlmostDisjoint). *Let X, Y be AlmostDisjoint over (A, B) with $n \geq 2$, and let M be a taut filling of $X+Y$. Then, writing $M_A = M \upharpoonright_{\subseteq A}$ and $M_B = M \upharpoonright_{\not\subseteq A}$,*

$$M = M_A + M_B, \quad \partial M_A = X, \quad \partial M_B = Y,$$

and M_A, M_B are themselves taut (`IsTaut.splits_full`).

Proof. Tautness enters precisely here, turning Theorem 2.3 into rigidity. Since M is taut, $\|M\| = \text{Zvol}(X+Y) = \text{Zvol } X + \text{Zvol } Y$, while $\|M\| \geq \|K_{A,p}M\| + \|K_{B,q}M\| \geq \text{Zvol } X + \text{Zvol } Y$. So all inequalities are equalities, with two consequences.

No double kill (`no_double_kill`): if some generator were retained by neither operator, the middle sum would be strictly smaller than $\|M\|$. Hence every generator is retained by exactly one side, and $K_{A,p}M, K_{B,q}M$ are *taut* fillings of X, Y .

No straddling simplex (`no_extreme_hybrid`). The three-part decomposition leaves, besides pure generators ($\subseteq A$ or $\subseteq B$), only “extreme hybrids”: a single vertex y_0 on the far side and none in $A \cap B$. If such hybrids existed, collect those through a fixed y_0 ; their faces opposite y_0 form a nonempty cycle L . That L is closed is *not* the $\mathbb{F}_2$ cut: a coefficient computation shows that ∂U carries no face through y_0 (such faces lie in neither X nor Y), and `bdry_lk_eq_zero` then converts this into $\partial L = 0$. So M would contain the complete cone $y_0 * L$, contradicting (T3). This is the only place $n \geq 2$ is used: the opposite faces need ≥ 2 vertices for L to be a genuine cycle.

Thus every generator of M is $\subseteq A$ or $\subseteq B$, so the literal restrictions give $M = M_A + M_B$. The identity $\partial M_A = X$ follows by applying $K_{A,p}$ to $\partial M_A + \partial M_B = X+Y$ (`kmap_eq_self` on the A -part, `kmap_eq_zero_of_closed` on the closed B -part); then $\partial M_B = Y$ by cancellation. Tautness of the pieces follows from $\|M_A\| + \|M_B\| = \|M\|$, additivity, and $\|M_A\| \geq \text{Zvol } X$, $\|M_B\| \geq \text{Zvol } Y$. $\square$

Remark (Why exactly AlmostDisjoint). The proof uses the separation hypothesis only through Lemma 2.2, which needs nothing beyond $|A \cap B| \leq n+1$. There is no appeal to any geometric or topological transversality; AlmostDisjointness is the exact combinatorial content that makes the cross terms vanish.

The rational corollary

The integer theory transfers to rational chains. A *rational chain* is a finitely supported $M: \{\text{simplices}\} \rightarrow \mathbb{Q}$ (`QChain`), with the same boundary (`Qbdry`) and cone (`Qcone`); its norm

$\text{Qnm} M = \sum_s |M(s)|$ and filling volume $\text{Qvol}(X) = \inf\{\text{Qnm} M : \partial M = X\}$ are real-valued. The Lean notion of rational tautness is *order-optimal*: M is `IsQTaut` if $\text{Qnm} M \leq \text{Qnm} N$ for every rational chain N with $\partial N = \partial M$ — norm-minimal among co-boundary competitors. This is the exact Lean definition; it does *not* assert that the infimum Qvol is attained, so no linear-programming or integrality theorem is needed.

Corollary 2.5 (Rational additivity and splitting). *Let X, Y be rational cycles, *AlmostDisjoint* over (A, B) (the same hypothesis bundle as Definition 1.3, now over $\mathbb{Q}$).*

- (i) *If $n \geq 1$ then $\text{Qvol}(X + Y) = \text{Qvol} X + \text{Qvol} Y$ (`Qvol_add_of_almost_disjoint_full`).*
- (ii) *If $n \geq 2$ and M is an order-optimal (`IsQTaut`) rational filling of $X + Y$, then $M = M_A + M_B$ with $\partial M_A = X$ and $\partial M_B = Y$, and each piece is again order-optimal (`IsQTaut.splits_full`).*

Proof. Clearing denominators: every rational chain scales by some $q \in \mathbb{N}$ to an integer chain (`exists_nat_smul_integral_three`). Part (i) scales the cycles X, Y and applies integer additivity (Theorem 2.3); part (ii) additionally scales the filling M and applies integer splitting (Theorem 2.4). Dividing back by q gives the rational statement in each case, order-optimality transferring because Qnm is homogeneous under positive scaling. $\square$

3 Taut fillings are AnyRootedStickerballs

We now prove the main theorem by induction on the number of tetrahedra, using Theorem 2.4 (in its 2-sphere “cut” form) to produce the smaller taut fillings. The same reduction first pays off a debt in the hypotheses: a taut filling of a 2-sphere is *automatically* simplicial, so “simplicial” is a conclusion, not an assumption.

Simpliciality is automatic

Theorem 3.1 (Simpliciality is automatic). *Let σ be a combinatorial 2-sphere, X the ± 1 cycle on σ , and M a taut filling with $\partial M = X$. Then every tetrahedron coefficient of M lies in $\{-1, 0, 1\}$ — M is simplicial (`taut_filling_is_simplicialChain`). No simpliciality is assumed.*

This discharges the only hypothesis of the main theorem that is not already part of “ M is a taut filling of the simplicial 2-sphere σ .” The proof is a minimal-counterexample deletion driven by a counting engine that produces removable tetrahedra *without* knowing the coefficients are ± 1 — the point at which a naive argument would be circular.

Lemma 3.2 (Disjoint eligible family, simpliciality-free). *Let σ have no degree-3 vertex, and let M be a taut filling whose boundary ∂M is the ± 1 cycle on σ and whose generators are all tetrahedra. Then for every vertex v there are at least $\deg_v(\partial M)$ generators of M , pairwise disjoint in their boundary faces, each sharing exactly two faces with ∂M in proper orientation — i.e. at least $\deg_v(\partial M)$ disjoint eligible tetrahedra (`exists_disjoint_eligible_family_noS`). (Under the unit boundary an eligible tetrahedron necessarily has coefficient ± 1 , `tet_coeff_eq_pm_one_of_eligible_unitOn`; this is what makes it removable below.)*

Proof. Call a triangle $\alpha \in \sigma$ a *same-sign support face* of a generator t if α is a face of t whose chain contribution carries the sign of ∂M there, i.e. $\partial M(\alpha) \cdot M(t) \cdot \text{sgn}(t, \alpha) > 0$ (`SameSignSupportFaceTet`). Every boundary triangle is a same-sign support face of *some* generator (`sameSignSupportFace_cover`); this — strictly weaker than “each boundary triangle lies on

a unique tetrahedron” — is the *only* geometric input, and it uses only that ∂M is the unit cycle and that generators are tetrahedra, *not* that their coefficients are ± 1 . Pick one such generator for each of the $|\sigma|$ boundary triangles; since no vertex has degree 3, each generator is picked by at most two triangles (`sharedFaces_card_le_two_of_noDegree3`).

Now count. Let $s = |\text{supp } M|$ be the number of generators and z the number of them with $|M(t)| \geq 2$. The norm budget $s + z \leq \|M\|$ (`support_card_add_nonUnitSupport_card_le_nrm`; each non-unit generator costs ≥ 2), tautness $\|M\| = \text{Zvol}(\partial M)$, and the degree–volume inequality $\text{Zvol}(\partial M) + \deg_v(\partial M) \leq \|\partial M\| = |\sigma|$ (`Zvol_add_deg_le`, `UnitOn.nrm_eq`) combine to

$$s + z + \deg_v(\partial M) \leq |\sigma|.$$

A map from the $|\sigma|$ triangles *into* the s generators with every fibre of size ≤ 2 therefore has at least $z + \deg_v(\partial M)$ *twofer* targets — generators picked by exactly two triangles — with pairwise-disjoint assigned pairs (`finset_twofer_fibers`: a fibred map with $s + k \leq |\sigma|$ has $\geq k$ size-2 fibres). Discard the $\leq z$ non-unit twofers; at least $\deg_v(\partial M)$ *unit* twofers remain. A unit twofer has coefficient ± 1 and two same-sign boundary faces, which on a unit tetrahedron are *properly oriented* (`sameSignFace_to_proper_of_unit`); it is therefore eligible, and disjoint assigned pairs make the family boundary-face disjoint. $\square$

Remark. The discard is the one place coefficients re-enter, and it is essential. Counting same-sign faces is insensitive to multiplicity — a generator of coefficient 2 still has each face once — so the twofer count runs with no simpliciality; but only a *unit* twofer is removable as an eligible tetrahedron. The norm budget pays the discard exactly: each non-unit generator contributes the ≥ 2 to $\|M\|$ that the count demanded as surplus.

Proof of Theorem 3.1. Strong induction on $\|M\|$, by the same trichotomy on σ as in the main induction below. If some generator t_0 has $|M(t_0)| \geq 2$ (`not_simplicialChain_iff_exists_natAbs_ge_two`), we produce a strictly smaller taut filling of a 2-sphere in which t_0 keeps its coefficient — still non-simplicial, contradicting the inductive hypothesis.

- *Base* ($|\sigma| \leq 4$): M is a single tetrahedron, necessarily of coefficient ± 1 , so there is no defect (`base_simplicialChain`).
- *Degree-3 vertex*: the capped cut (Lemma 2.1) splits $M = M_L + M_R$ into two strictly smaller taut fillings of the two capped spheres (`taut_splits_for_capped_cut`), and t_0 lies on one side carrying its full coefficient.
- *No degree-3 vertex*: applied at any vertex (every vertex of a 2-sphere has $\deg_v(\partial M) \geq 3$), Lemma 3.2 supplies at least three eligible tetrahedra — in particular two distinct ones, so one is $u \neq t_0$. Remove u ; t_0 persists with coefficient $M(t_0)$ (`removeTet_apply_ne`). If u ’s flip edge is absent, $M - u$ is a taut filling of the flipped 2-sphere (`isSphere2_flipBoundary_of_eligible`, `isTaut_removeTet`); if present, $M - u$ splits along that edge into two taut sub-fillings on smaller spheres (`flipEdgePresent_side_sets_noS`, `flipEdgePresent_side_algebra_noS`), and t_0 lies on one.

In every case the smaller filling is simplicial by induction, forcing $|M(t_0)| \leq 1$ — a contradiction. Hence M is simplicial. $\square$

The induction

With simpliciality established (Theorem 3.1) we use it freely below: the Lean lemmas `taut_isPseudomanifold` and `theorem3_clean` carry `SimplicialChain M` as an explicit hypothesis, discharged once and for all by that theorem. First, the engine that gives the filling its 3-dimensional shape.

Theorem 3.3 (Taut fillings are pure and triangle-bounded). *Let σ be a combinatorial 2-sphere, X the ± 1 cycle on σ (`UnitOn X  $\sigma$`), and M a taut filling with $\partial M = X$. Then every generator of M is a tetrahedron and every triangle lies in at most two of them (`taut_isPseudomanifold`, simpliciality from Theorem 3.1; purity from `IsTaut.dim_pure`).*

Proof. Purity is (T2): X is 2-dimensional, so a taut filling is 3-dimensional. Triangle-boundedness is a strong induction on the number of tetrahedra, by the trichotomy used again below. *Base* (σ has ≤ 4 vertices): M is a single tetrahedron. *Degree-3 vertex*: cut its star (Lemma 2.1), apply the inductive hypothesis to the smaller filling, re-glue. *No degree-3 vertex*: take two disjoint tetrahedra e, u each meeting the boundary in exactly two faces (`aleph_disjoint_eligible_pair`), remove e , apply the inductive hypothesis, re-glue; the key point is the *triangle rule-out* (`faceCount_removeTet_sharedFace_eq_zero`): a shared face of e cannot reappear after removing e , since the disjoint witness u would otherwise force that triangle's incidence to 3 in the inductively triangle-bounded filling $M - u$. $\square$

Theorem 3.4 (Main). *Let σ be a combinatorial 2-sphere, X the ± 1 cycle on σ , and M a taut filling with $\partial M = X$. Then the set of tetrahedra of M is an `AnyRootedStickerball` with boundary σ (`taut_filling_is_anyrootedStickerball`). Simpliciality is not assumed: the endpoint derives it internally by Theorem 3.1.*

By Theorem 3.3 the support consists of tetrahedra; $\sigma \neq \emptyset$ gives nonempty boundary. The content is that every tetrahedron is a root:

Theorem 3.5. *Under the hypotheses of Theorem 3.4, for every tetrahedron $t_\star$ of M there is a clean shelling of M beginning with $t_\star$ (`theorem3_clean`, simpliciality from Theorem 3.1, via the induction skeleton `theorem3_core_clean` with cases `base_free_clean`, `deg3_step_clean`, `prime_step_clean`).*

Proof. Strong induction on the number N of tetrahedra of M . The inductive invariant is exactly the `AnyRooted` statement: a clean shelling beginning at *any* prescribed $t_\star$. The cases are a trichotomy on σ . In each non-base case the cut (Lemma 2.1) supplies strictly smaller taut fillings of smaller spheres to which the inductive hypothesis applies — by Theorem 2.4 itself in the degree-3 case, and by a separate side-decomposition resting on the same cut in the prime case.

Base (σ has ≤ 4 vertices). Then σ is a tetrahedron boundary and M is a single tetrahedron T ; the one-term shelling starts at T and is vacuously clean.

Degree-3 vertex v . The link of v is a triangle γ , and the star of v is one tetrahedron $T = v * \gamma$. Since $|\sigma| \geq 5$, γ is a non-face triangle whose edges lie in σ , so by Lemma 2.1 replacing the three triangles at v by γ yields a 2-sphere σ' with one fewer vertex. Restricting M to the non-star side is a *direct instance of Theorem 2.4*, with the two capped vertex-sets as A, B (their overlap is exactly γ , so $|A \cap B| = n+1 = 3$): the lemma `taut_splits_for_capped_cut` invokes `IsTaut.splits` verbatim. It produces a taut filling M' of σ' with strictly fewer tetrahedra, so by induction M' is an `AnyRootedStickerball`. Two facts let T be glued cleanly: γ is a boundary triangle of M' lying in a *unique* remainder tetrahedron t_0 (tautness pins it, `faceCount( $\gamma$ ) = 1`; `degree3_anchor`), and v is

a vertex absent from M' (`degree3_apex_notMem_*`), so adjoining T along the single face γ satisfies (a)–(c3) (the apex v is new, so its link certificates are vacuous). Now assemble, given $t_\star$:

- if $t_\star \neq T$: take a clean shelling of M' rooted at $t_\star$ (induction) and append T last;
- if $t_\star = T$: start with T , then a clean shelling of M' rooted at the anchor t_0 (induction), which glues to T along γ .

Either way M has a clean shelling rooted at $t_\star$.

No degree-3 vertex ($|\sigma| \geq 5$). Pick a tetrahedron e of M with exactly two faces on σ (its existence is a count: with no degree-3 vertex there are more boundary triangles than tetrahedra; `exists_eligibleTet`). Let f_3, f_4 be its two interior faces and $f_3 \cap f_4$ its *flip edge*. Remove e .

- *Flip edge absent*. Exposing f_3, f_4 is a 2–2 flip of the boundary: the new boundary is again a 2-sphere σ' on the same vertices (`isSphere2_flipBoundary_of_eligible`) and $M - e$ is a taut filling of it with fewer tetrahedra. By induction $M - e$ is an `AnyRootedStickerball`; re-gluing e (two shared faces) is a clean glue step, its shared faces clean by the rule-out of Theorem 3.3.
- *Flip edge present*. Then $f_3 \cap f_4 \in \sigma$, and σ separates along this edge into two 2-spheres σ_1, σ_2 ; correspondingly $M - e$ splits along this edge into two taut sub-fillings M_1, M_2 with fewer tetrahedra (`flipEdgePresent_side_sets`, `flipEdgePresent_side_algebra`). This is a decomposition of the same filter-by-side shape as Theorem 2.4, but a *separate* argument: it is built directly on the cut (Lemma 2.1/`exists_cut`), not through `IsTaut.splits`. By induction each is an `AnyRootedStickerball`; concatenate their clean shellings — rooting the second at the tetrahedron adjacent to the bridge — and insert e to bridge them (`glueStep_bridge_left`, `glueStep_bridge_right`).

In both sub-cases, choosing the root of the smaller piece(s) appropriately (and, if $t_\star = e$, starting with e) yields a clean shelling of M rooted at $t_\star$.

This proves the inductive invariant for M , completing the induction. □

Proof of Theorem 3.4. Theorem 3.5 gives a clean shelling rooted at each tetrahedron; with nonempty boundary this is the `AnyRootedStickerball` conclusion. Each re-gluing above is a clean glue step, so the link certificates (c1)–(c3) hold at every step; the assembled shelling is therefore a genuine clean shelling and the rooted structure is preserved through the assembly. □

Remark (Where each ingredient is used). The recursion always lands on strictly smaller taut fillings of strictly smaller spheres, so the induction terminates. Tautness is used (i) to make every cut-restriction again taut, so the inductive hypothesis applies (Theorem 2.4 for the degree-3 cut, the analogous side-decomposition for the prime case); (ii) through Theorem 3.3 to keep every intermediate complex triangle-bounded, which the prime rule-out needs; and (iii) to pin boundary triangles into unique tetrahedra, giving the gluing anchors. The rooted structure is preserved because every assembly step is a *clean* glue step.

4 Taut fillings are flag complexes

A simplicial complex is a *flag complex* when every clique of its 1-skeleton already spans a face: if a set of vertices has all of its edges present, the whole set is a simplex. A taut filling inherits this.

Definition 4.1 (Simplex, clique, flag complex). For a set τ of tetrahedra, a vertex set s is a *simplex* of τ if $s = \emptyset$ or $s \subseteq t$ for some $t \in \tau$ (`SimplexOf`). It is a *clique* of the 1-skeleton if $s \subseteq \text{verts } \tau$ and every 2-subset of s is a simplex (`CliqueInOneSkeleton`). Then τ is a *flag complex* (`IsFlagComplex`) if every clique is a simplex.

Theorem 4.2 (Flag complex). *Let σ be a combinatorial 2-sphere, X the ± 1 cycle on σ , and M a taut filling with $\partial M = X$. Then the tetrahedra of M form a flag complex (`taut_filling_is_flagComplex`). Simplicity is not assumed: the public endpoint derives it internally from Theorem 3.1, exactly as the stickerball endpoint does.*

The proof isolates three forbidden “taboo” configurations and rules each out.

Lemma 4.3 (No taboo $\Rightarrow$ flag). *Call three vertices an empty K_3 if they carry all their edges yet do not span a simplex; four vertices an empty K_4 likewise; and five vertices a K_5 of edges if they carry all ten edges. If τ has none of the three, it is a flag complex (`NoTaboo.to_flag`): a clique of size 0, 1, 2 is a simplex outright, size 3 or 4 would be an empty K_3 or K_4 , and size ≥ 5 contains a K_5 of edges.*

Proof of Theorem 4.2. By Lemma 4.3 it suffices to rule out the three taboos for a taut filling.

No empty K_3 or K_4 (`no_emptyK3K4_of_taut`): a strong induction on $\|M\|$ by the same base / degree-3 / prime trichotomy as §3. Each reduction (a degree-3 peel or an eligible-tet flip) lands on a strictly smaller taut filling; an empty K_3 or K_4 would persist into it, contradicting the inductive hypothesis.

No K_5 of edges (`no_k5Clique_of_no_emptyK4_taut`). This is the combinatorial substitute for the paper’s “ $S^3 \not\subseteq B^3$.” Suppose five vertices s carry all ten edges. Because there is no empty K_4 , some 4-subset of s is an actual tetrahedron of M (`fourSubset_mem_support`), so the subchain $U = M \upharpoonright_{\subseteq s}$ is nonzero. But with triangle-boundedness (Theorem 3.3) every triangle inside s is interior, so $\partial U = 0$ (`bdry_filter_subset_eq_zero`); and a subchain of a taut chain is taut (`IsTaut.subChain`), forcing $\|U\| = \text{Zvol}(0) = 0$, i.e. $U = 0$ — a contradiction. $\square$

Remark. The K_5 step is where “a taut filling has no nonzero closed subchain” replaces the paper’s appeal to $S^3 \not\subseteq B^3$; the project carries no topology, and this combinatorial substitute is what is Lean-proven. The internal lemmas (`theorem4_flag`, `no_emptyK3K4_of_taut`) still take `SimplicialChain M` as data — threaded through their own induction — but the public endpoint `taut_filling_is_flagComplex` supplies it from Theorem 3.1, so the flag-complex theorem itself carries *no* simpliciality hypothesis (its signature is `IsSphere2  $\sigma \rightarrow \text{UnitOn } X \sigma \rightarrow \partial X = 0 \rightarrow \partial M = X \rightarrow \text{IsTaut } M \rightarrow \text{IsFlagComplex } M.\text{support}$`).

5 Lean correspondence and status

The development builds successfully; a search of `lean/Taut` finds no `sorry`, `admit`, `native_decide`, `axiom`, or `unsafe`. Each theorem below reduces to the three foundational axioms `propext`, `Classical.choice`, `Quot.sound` only.

Result here	Lean identifier	Status
Lem 2.1 ($H_1 = 0$, cut)	<code>range_bd2_eq_ker_bd1, exists_cut</code>	fully proved
Thm 2.3 (additivity)	<code>Zvol_add_of_almost_disjoint_full</code>	fully proved
Thm 2.4 (splitting)	<code>IsTaut.splits_full</code>	fully proved
Thm 3.1 (simpliciality automatic)	<code>taut_filling_is_simplicialChain</code>	fully proved
Thm 3.3 (pure, triangle-bounded)	<code>taut_isPseudomanifold</code>	fully proved
Thm 3.5 (rooted at each tet)	<code>theorem3_clean</code>	fully proved
Thm 3.4 (AnyRootedStickerball)	<code>taut_filling_is_anyrootedStickerball</code>	fully proved
Thm 4.2 (flag complex)	<code>taut_filling_is_flagComplex</code>	fully proved
Cor 2.5(i) (rational additivity)	<code>Qvol_add_of_almost_disjoint_full</code>	fully proved
Cor 2.5(ii) (rational splitting)	<code>IsQTaut.splits_full</code>	fully proved

Definitions weaker than their classical namesakes.

- *Taut* (`IsTaut`) is combinatorial ℓ^1 -minimality (Definition 1.1), not a geometric condition.
- *Triangle-bounded* (`IsPseudomanifold` in the development) means only “every triangle in at most two tetrahedra.” It is *not* a full pseudomanifold; normality (link-connectedness) is established separately, via the clean shelling (see §6). Theorem 3.3 proves the triangle bound only.
- *Clean shelling* (`IsCleanShelling`) carries the link certificates (c1)–(c3) as data; this is what makes it certify a clean 3-complex, and is stronger than a boundary-only shelling.

6 Relation to the classical statement; remaining bridges

The conclusion of Theorem 3.4 is stated in Lean-backed terms (`AnyRootedStickerball`). Two questions separate this from a classical “shellable normal 3-ball” statement.

Proved bridge: `AnyRootedStickerball` $\Rightarrow$ **normal 3-pseudomanifold**. The development proves that a stickerball is a clean 3-complex:

`IsStickerball.isClean3Complex`: `stickerball` $\implies$ `pure` $\wedge$ `triangle-bounded` $\wedge$ `Normal3`,

where `Normal3` is *edge-link connected* and *vertex-link connected* (`IsAnyrootedStickerball.isClean3Complex` for the `AnyRooted` form). This is proved by iterating the single-insertion lemma `clean3Complex_insert` along the clean shelling; the three certificates (c1)–(c3) of each glue step supply exactly the triangle-count, edge-link, and vertex-link preservation (`isPseudomanifold_insert`, `edgeLinkConnected_insert`, `vertexLinkConnected_insert`). Thus pseudomanifoldness (triangle bound), edge-link connectedness, vertex-link connectedness, and nonempty boundary are all *Lean-proved* consequences of the conclusion; the corresponding corollaries are `taut_filling_is_clean3Complex` and `taut_filling_is_shellable`.

Missing bridge: `stickerball` $\Rightarrow$ **topological/PL 3-ball**. The statement “a clean-shellable normal 3-pseudomanifold with nonempty boundary is (PL-)homeomorphic to the 3-ball” is the classical shelling theorem of Danaraj–Klee / Björner. This bridge is **not Lean-proven here**: no theorem from `IsStickerball` to a topological or PL ball appears in the development (it is recorded only as a docstring remark). Any classical reading of Theorem 3.4 as “ M triangulates a 3-ball” must invoke that external theorem explicitly. In the present, fully formalized terms the conclusion is exactly: *the tetrahedra of a taut filling of a combinatorial 2-sphere form an AnyRootedStickerball — a clean-shellable normal 3-pseudomanifold, with boundary the given sphere, that can be clean-shelled starting from any tetrahedron.*

Hypotheses to carry into a paper statement. The boundary cycle is ± 1 on σ (`Unit0n`); the filling's simpliciality is *derived, not assumed* (Theorem 3.1, `taut_filling_is_simplicialChain`), so the only remaining qualifier on M is tautness. Splitting (Theorem 2.4) needs dimension $n \geq 2$ while additivity (Theorem 2.3) needs only $n \geq 1$; `AlmostDisjointness` is $|A \cap B| \leq n + 1$; the ambient V is infinite; and `IsSphere2` includes the Euler relation $V + F = E + 2$ as part of its definition, from which $H_1 = 0$ (Lemma 2.1) is derived rather than assumed.